

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from google.colab import files
```

```
import pandas as pd

df = pd.read_csv("Iris.csv")
df.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
#   Column              Non-Null Count  Dtype
---  ---
0    Id                  150 non-null   int64
1    SepalLengthCm       150 non-null   float64
2    SepalWidthCm        150 non-null   float64
3    PetalLengthCm       150 non-null   float64
4    PetalWidthCm        150 non-null   float64
5    Species              150 non-null   object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

```
df.describe()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

```
df.isnull().sum()
```

```
0
Id      0
SepalLengthCm  0
SepalWidthCm  0
PetalLengthCm  0
PetalWidthCm  0
Species     0

dtype: int64
```

```
df.columns
```

```
Index(['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',  
      'Species'],  
      dtype='object')
```

```
df = df.drop('Id', axis=1)
```

```
df.columns
```

```
Index(['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',  
      'Species'],  
      dtype='object')
```

```
X = df.drop('Species', axis=1)  
y = df['Species']
```

```
X.head()
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

```
y.head()
```

	Species
0	Iris-setosa
1	Iris-setosa
2	Iris-setosa
3	Iris-setosa
4	Iris-setosa

```
dtype: object
```

```
from sklearn.model_selection import train_test_split  
  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)
```

```
from sklearn.ensemble import RandomForestClassifier  
  
model = RandomForestClassifier()  
model.fit(X_train, y_train)
```

```
▼ RandomForestClassifier ⓘ ?  
RandomForestClassifier()
```

```
y_pred = model.predict(X_test)
```

```
from sklearn.metrics import accuracy_score  
  
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 1.0
```

```
from sklearn.metrics import classification_report, confusion_matrix  
  
print(confusion_matrix(y_test, y_pred))  
print(classification_report(y_test, y_pred))
```

```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	10
Iris-versicolor	1.00	1.00	1.00	9
Iris-virginica	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30